

MM74C906

Hex Open Drain N-Channel Buffers

General Description

The MM74C906 buffer employs monolithic CMOS technology in achieving open drain outputs. The MM74C906 consists of six inverters driving six N-channel devices. The open drain feature of these buffers makes level shifting or wire AND and wire OR functions by just the addition of pull-up or pull-down resistors. All inputs are protected from static discharge by diode clamps to V_{CC} and to ground.

Features

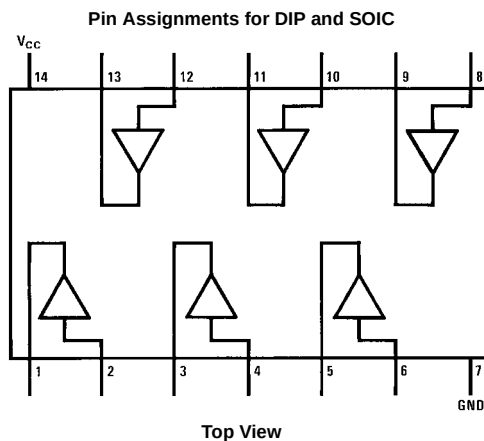
- Wide supply voltage range: 3V to 15V
- Guaranteed noise margin: 1V
- High noise immunity: $0.45 V_{CC}$ (typ.)
- High current sourcing and sinking open drain outputs

Ordering Code:

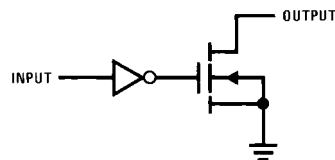
Order Number	Package Number	Package Description
MM74C906M (Note 1)	M14A	14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150" Narrow
MM74C906N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300" Wide

Note 1: Devices also available in Tape and Reel. Specify by appending the suffix letter "X" to the ordering code.

Connection Diagram



Logic Diagram



Absolute Maximum Ratings(Note 2)

Voltage at Any Input Pin	−0.3V to $V_{CC} + 0.3V$
Voltage at Any Output Pin	
Operating Temperature Range	−40°C to +85°C
Storage Temperature Range	−65°C to +150°C
Power Dissipation	
Dual-In-Line	700 mW
Small Outline	500 mW
Operating V_{CC} Range	3V to 15V
Absolute Maximum V_{CC}	18V
Lead Temperature (T_L)	
(Soldering, 10 seconds)	260°C

Note 2: "Absolute Maximum Ratings" are those values beyond which the safety of the device cannot be guaranteed. Except for "Operating Temperature Range" they are not meant to imply that the devices should be operated at these limits. The table of "Electrical Characteristics" provides conditions for actual device operation.

DC Electrical Characteristics

Min/Max limits apply across temperature range unless otherwise noted

Symbol	Parameter	Conditions	Min	Typ	Max	Units
CMOS TO CMOS						
$V_{IN(1)}$	Logical "1" Input Voltage	$V_{CC} = 5V$ $V_{CC} = 10V$	3.5 8.0			V V
$V_{IN(0)}$	Logical "0" Input Voltage	$V_{CC} = 5V$ $V_{CC} = 10V$			1.5 2	V V
$I_{IN(1)}$	Logical "1" Input Current	$V_{CC} = 15V, V_{IN} = 15V$		0.005	1	μA
$I_{IN(0)}$	Logical "0" Input Current	$V_{CC} = 15V, V_{IN} = 0V$	−1.0	−0.005		μA
I_{CC}	Supply Current	$V_{CC} = 15V$, Output Open		0.05	15	μA
	Output Leakage	$V_{CC} = 4.75V, V_{IN} = V_{CC} - 1.5V$ $V_{CC} = 4.75V, V_{OUT} = 18V$		0.005	5	μA
CMOS/LPTTL INTERFACE						
$V_{IN(1)}$	Logical "1" Input Voltage	$V_{CC} = 4.75V$	$V_{CC} - 1.5V$			V
$V_{IN(0)}$	Logical "0" Input Voltage	$V_{CC} = 4.75V$			0.8	V
OUTPUT DRIVE CURRENT						
		$V_{CC} = 4.75V, V_{IN} = 1V + 0.1 V_{CC}$ $V_{CC} = 4.75V, V_{OUT} = 0.5V$ $V_{CC} = 4.75V, V_{OUT} = 1.0V$	2.1 4.2	8.0 12.0		mA mA
		$V_{CC} = 10V, V_{IN} = 2V$ $V_{CC} = 10V, V_{OUT} = 0.5V$ $V_{CC} = 10V, V_{OUT} = 1V$	4.2 8.4	20 30		mA mA

AC Electrical Characteristics (Note 3)

$T_A = 25^\circ\text{C}$, $C_L = 50\text{ pF}$, unless otherwise specified

Symbol	Parameter	Conditions	Min	Typ	Max	Units
t_{pd}	Propagation Delay Time to a Logical "0"	$V_{CC} = 5.0\text{V}$, $R = 10\text{k}$ $V_{CC} = 10\text{V}$, $R = 10\text{k}$			150 75	ns ns
t_{pd}	Propagation Delay Time to a Logical "1"	$V_{CC} = 5.0\text{V}$ (Note 4) $V_{CC} = 10\text{V}$ (Note 4)			$150 + 0.7 RC$ $75 + 0.7 RC$	ns ns
C_{IN}	Input Capacitance	(Note 5)		5.0		pF
C_{OUT}	Output Capacity	(Note 5)		20		pF
C_{PD}	Power Dissipation Capacity	(Note 6) Per Buffer		30		pF

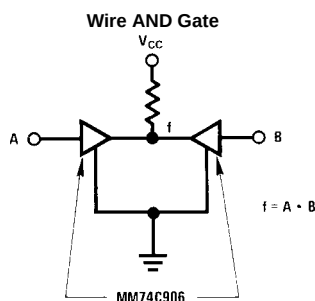
Note 3: AC Parameters are guaranteed by DC correlated testing.

Note 4: "C" used in calculating propagation includes output load capacity (C_L) plus device output capacity (C_{OUT}).

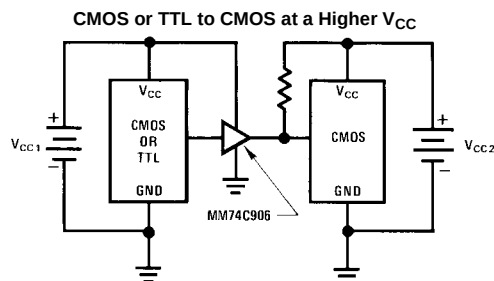
Note 5: Capacitance is guaranteed by periodic testing.

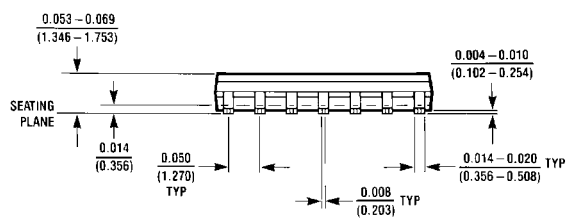
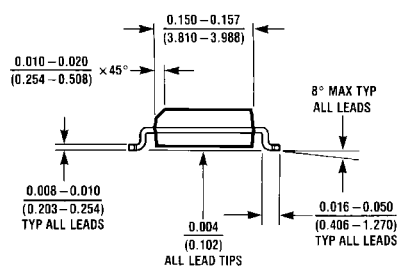
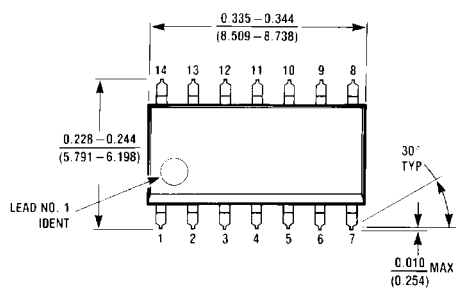
Note 6: C_{PD} determines the no load AC power consumption of any CMOS device. For complete explanation see Family Characteristics Application Note, AN-90. (Assumes outputs are open).

Typical Applications



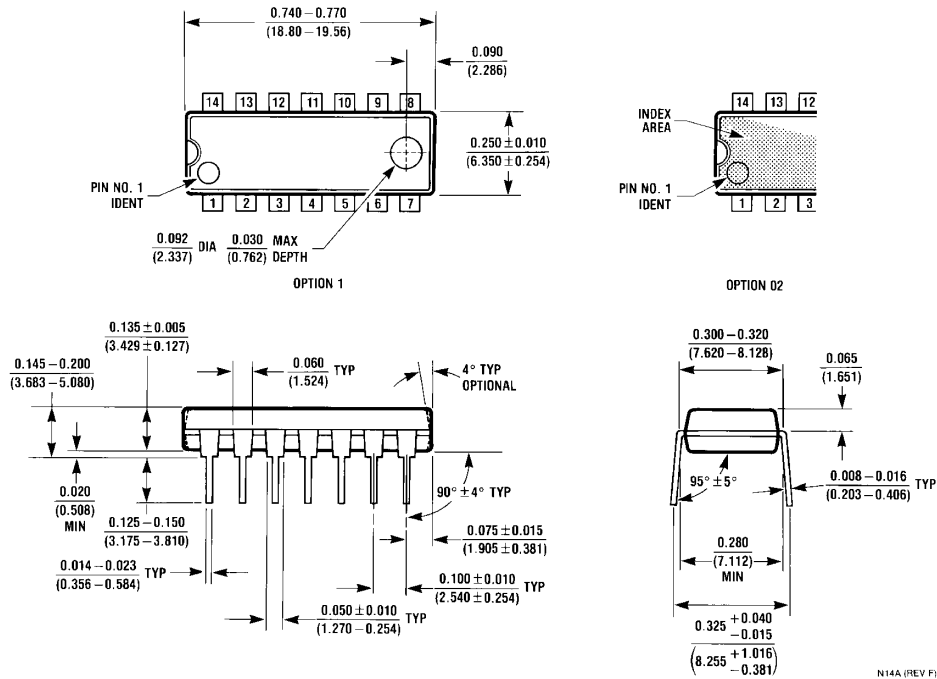
Note: Can be extended to more than 2 inputs.





**14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150" Narrow
Package Number M14A**

Physical Dimensions inches (millimeters) unless otherwise noted (Continued)



14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300" Wide
Package Number N14A

Fairchild does not assume any responsibility for use of any circuitry described, no circuit patent licenses are implied and Fairchild reserves the right at any time without notice to change said circuitry and specifications.

LIFE SUPPORT POLICY

FAIRCHILD'S PRODUCTS ARE NOT AUTHORIZED FOR USE AS CRITICAL COMPONENTS IN LIFE SUPPORT DEVICES OR SYSTEMS WITHOUT THE EXPRESS WRITTEN APPROVAL OF THE PRESIDENT OF FAIRCHILD SEMICONDUCTOR CORPORATION. As used herein:

1. Life support devices or systems are devices or systems which, (a) are intended for surgical implant into the body, or (b) support or sustain life, and (c) whose failure to perform when properly used in accordance with instructions for use provided in the labeling, can be reasonably expected to result in a significant injury to the user.
2. A critical component in any component of a life support device or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.

www.fairchildsemi.com

Mouser Electronics

Authorized Distributor

Click to View Pricing, Inventory, Delivery & Lifecycle Information:

[Fairchild Semiconductor:](#)

[MM74C906M_Q](#)